

Rail Fence Cipher

Aim

To implement the Rail Fence cipher, a transposition cipher that writes the text in a zig-zag pattern across a number of “rails” (rows), then reads the rails off one after another to form the ciphertext. It only reorders characters, so every character (including spaces, punctuation, and case) is preserved.

Algorithm

```
1 Function Encrypt(plain_text, rails):
2   if rails ≤ 1 then
3     return plain_text                                // nothing to fold
4   lines ← array of rails empty strings
5   rail ← 0
6   dir ← +1
7   foreach character ch in plain_text do
8     append ch to lines[rail]
9     if rail = 0 then
10      dir ← +1
11     if rail = rails - 1 then
12      dir ← -1
13     rail ← rail + dir                                // bounce between top and bottom rail
14 return concatenation of all lines
```

```
1 Function Decrypt(cipher_text, rails):
2   if rails ≤ 1 then
3     return cipher_text
4   rail_of ← rail index for each position                // re-walk the same zig-zag
5   lines ← slice cipher_text into rails, taking each rail's count in order
6   idx ← array of rails zeros
7   plain ← empty string of same length
8   foreach position i following the zig-zag do
9     rail ← rail_of[i]
10    plain[i] ← lines[rail][idx[rail]]
11    idx[rail] ← idx[rail] + 1
12 return plain
```

Output

Encrypt — plaintext HELLOWORLD, rails 3

Writing the letters in a zig-zag over 3 rails (a dot marks an empty cell):

```
rail 0:  H . . . O . . . L .
rail 1:  . E . L . W . R . D
rail 2:  . . L . . . O . . .
```

Each character lands on a rail as the path bounces down then up:

Char	Index	Rail
H	0	0
E	1	1
L	2	2
L	3	1
O	4	0
W	5	1
O	6	2
R	7	1
L	8	0
D	9	1

Reading the rails top to bottom: rail 0 = HOL, rail 1 = ELWRD, rail 2 = LO.

Result: HELLOWORLD -> HOLELWRDLO

Decrypt — ciphertext HOLELWRDLO, rails 3

Re-walking the zig-zag shows each rail holds a known number of characters, so the ciphertext is sliced back into the rails first:

Rail	Count	Slice of HOLELWRDLO
0	3	HOL
1	5	ELWRD
2	2	LO

Then the zig-zag is followed again, taking the next unused character from the rail that each position belongs to (0,1,2,1,0,1,2,1,0,1), rebuilding the text.

Result: HOLELWRDLO -> HELLOWORLD